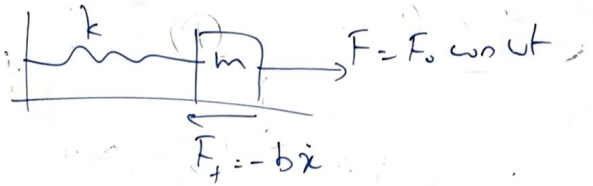
21.7.21

Waves Lec 16 — Power absorption in driven osc.



$$\ddot{x} + \gamma\dot{x} + \omega_0^2 x = \frac{F_0}{m}\cos\omega t$$

$$x = A\cos(\omega t - \delta)$$

$$(\omega_0^2 - \omega^2)A + i\gamma\omega A = \frac{F_0}{m}(\cos\delta + i\sin\delta)$$

$$A = \frac{f_0}{m}\frac{1}{\left[(\omega_0^2-\omega^2)^2 + (\gamma\omega)^2\right]^{1/2}}$$

$$v = \dot{x} = -\omega A\sin(\omega t - \delta)$$

$$= v_0(\omega)\sin(\omega t - \delta)$$

$$v_0(\omega) = \omega A(\omega)$$

$$P(t) = Fv$$

$$= -F_0 v_0(\omega)\cos\omega t\sin(\omega t - \delta)$$

$$\bar{P} = \langle P(t)\rangle = -F_0 v_0(\omega)\langle\cos\omega t\sin(\omega t - \delta)\rangle$$

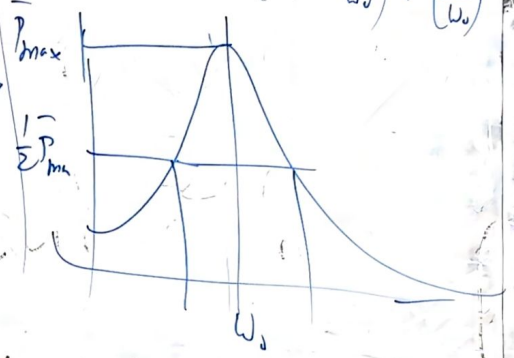
$$= \frac{1}{2}F_0 v_0(\omega)\sin\delta$$

$$= \frac{1}{2}v_0(\omega)\, m\omega A\gamma$$

$$\bar{P} = \frac{1}{2}m\omega^2 A^2\gamma$$

$$= \frac{F_0^2\gamma}{2m}\frac{\omega^2}{(\omega_0^2-\omega^2)^2 + (\gamma\omega)^2}$$

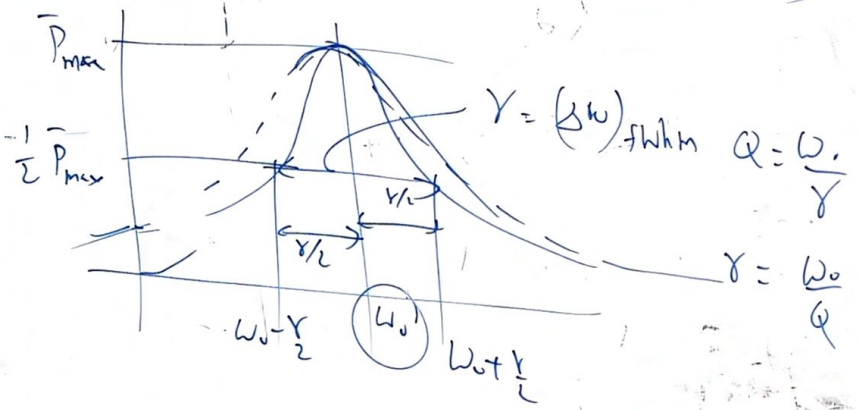
$$= \frac{F_0\gamma}{2m\omega_0^2}\frac{1}{\left(\frac{\omega_0}{\omega} - \frac{\omega}{\omega_0}\right)^2 + \left(\frac{\gamma}{\omega_0}\right)^2}$$



cont'd.

$$\bar{P}(\omega) \approx \frac{F_0^2 \gamma}{2m\omega_0^2} \cdot \frac{1}{4\frac{(\omega-\omega_0)^2}{\omega_0^2} + \frac{\gamma^2}{\omega_0^2}}$$

$$= \frac{F_0^2 \gamma}{8m} \left[\frac{1}{(\omega-\omega_0)^2 + \left(\frac{\gamma}{2}\right)^2} \right]$$



$\gamma = (\delta\omega)_{FWHM}$ $\quad Q = \frac{\omega_0}{\gamma}$

$\gamma = \frac{\omega_0}{Q}$

Resonance width $= \frac{\omega_0}{Q}$

Free damped osc. $\quad E(t) \approx E_0 e^{-\gamma t}$

$$\frac{dE}{dt} = -\gamma E \Rightarrow \frac{\Delta E}{\Delta t} \approx -\gamma E_0$$

$$\frac{\Delta E}{E} \approx -\gamma \Delta t$$

$$\left|\frac{\Delta E}{E}\right| \sim \gamma T = \gamma \cdot \frac{2\pi}{\omega_0} = 2\pi$$

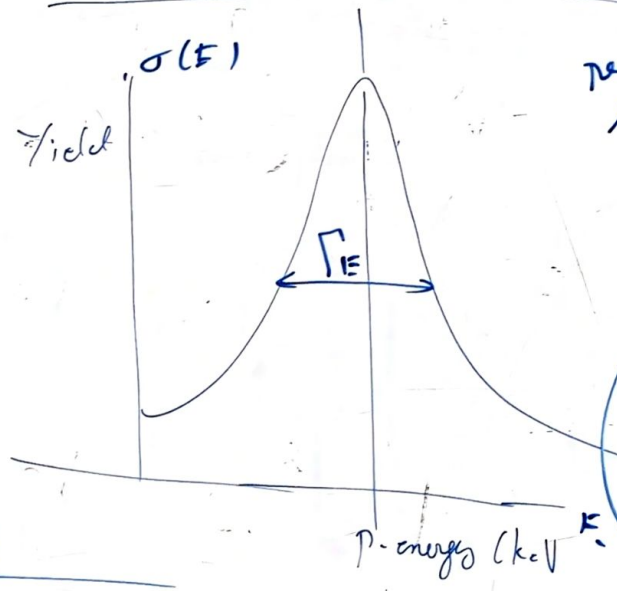
$$Q = \frac{2\pi E}{\Delta E(\text{cycle})}$$

21.7.21

Power Lec 16 — Power absorption in driven osc. 1

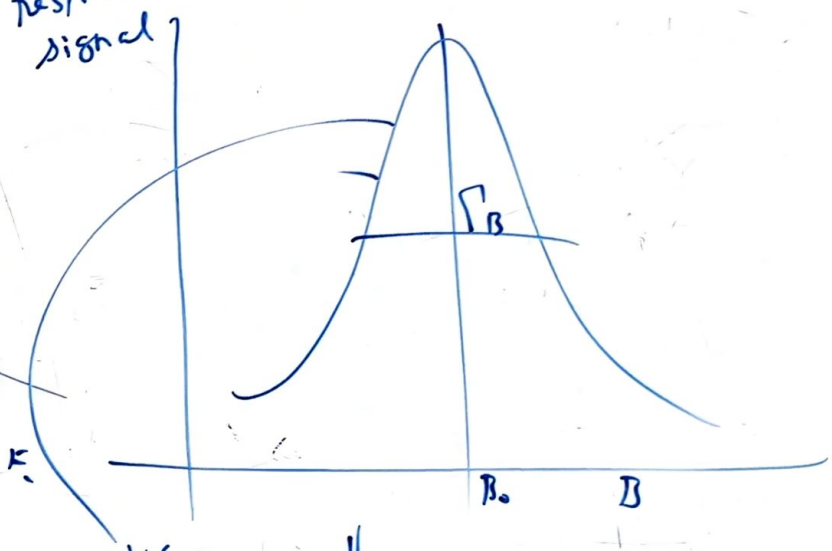
Resonance in the yield of γ-rays

$$p + {}^{19}F \longrightarrow {}^{20}Ne^{*} + \gamma$$



$$\sigma(E) = \frac{\sigma(E_0)}{\frac{(E-E_0)^2}{4\Gamma^2} + 1}$$

NMR



$$V(B) = \frac{V_0}{1 + \frac{(B-B_0)^2}{4\Gamma_B^2}}$$

Superposition Principle

Linear ODE such as

$$\ddot{x} + \gamma\dot{x} + \omega_0^2 x = 0$$

* If $x_1(t)$, $x_2(t)$ are solns.

Then $x = c_1 x_1(t) + c_2 x_2(t)$ is also a soln.

$$\ddot{x} + \gamma\dot{x} + \omega_0^2 x = 0$$

* If $x_1(t)$ is soln with $x_1(0) = x_{10}$, $\dot{x}_1(0) = v_{10}$

$x_2(t)$ is " " $x_2(0) = x_{20}$, $\dot{x}_2(0) = v_{20}$

Then, if $x(0) = x_{10} + x_{20}$, $\dot{x}(0) = v_{10} + v_{20}$

$x(t) = x_1(t) + x_2(t)$

$$\ddot{x}_1 = -k_1 x_1 - k_2 x_1^2 + \cdots$$

$$\ddot{x}_2 = -k_1 x_2 - k_2 x_2^2 + \cdots$$

$x = x_1 + x_2$

$\ddot{x} \stackrel{?}{=} -k_1 x - k_2 x^2$ ✗ Verify

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